

Why CPE removal buys less on 11ax/be than on 11ac/n

wifitrx phase-noise / CPE study — 40 MHz, shipped WiFi 7 LO profile, single LO

1. What CPE removal is: one number per symbol

Within one OFDM symbol of FFT length T , the received phase noise $\exp(j\varphi(t))$ becomes, after the DFT,

$$Y_k = J_0 X_k + \sum_{m \neq k} J_{k-m} X_m, \quad J_0 = \frac{1}{T} \int_0^T e^{j\varphi(t)} dt \approx e^{j\bar{\varphi}}$$

The sum over $m \neq k$ is the inter-carrier interference (ICI). CPE removal multiplies each symbol by $\exp(-j\bar{\varphi})$: it subtracts the symbol-mean phase $\bar{\varphi}$, a single number per symbol. Whatever remains, $\varphi(t) - \bar{\varphi}$, turns into ICI, and no single rotation can touch it.

Panel (1) of the figure (page 3): the grey trace is one LO phase realization; the green staircase is the per-symbol mean for 3.2 μs symbols, the red one for 12.8 μs . The red staircase follows the wander four times more coarsely, and the area between the grey trace and the red staircase is the ICI that 11ax keeps.

2. Taking the symbol mean is a sinc² weight in frequency

The symbol mean is a rectangular-window average, whose squared frequency response is

$$|W(f)|^2 = \text{sinc}^2(fT) = \left[\frac{\sin(\pi fT)}{\pi fT} \right]^2$$

so the phase-noise power is split by frequency into a removable part and a part left behind:

$$\sigma_{\text{CPE}}^2 = \int S_{\varphi}(f) \text{sinc}^2(fT) df \quad (\text{removable}), \quad \sigma_{\text{ICI}}^2 = \int S_{\varphi}(f) [1 - \text{sinc}^2(fT)] df \quad (\text{left behind})$$

Solving $\text{sinc}^2(fT) = 1/2$ gives the hand-over frequency:

$$f_{-3\text{dB}} = \frac{0.443}{T}$$

	T	f_{-3dB}
11ac/n	3.2 μs	138 kHz
11ax/be	12.8 μs	35 kHz

A 4× longer symbol makes the band CPE can act on 4× narrower. This is pure geometry and has nothing to do with the LO. The two S-shaped curves in panel (2) are the two weights; the shaded areas are $S_{\varphi}(f) \cdot \text{sinc}^2(fT)$, the slice of the spectrum each standard actually removes.

3. How much gets removed depends on where the LO's noise sits

The shipped LO profile has a flat plateau S_0 (−104.1 dBc/Hz) from 10 to 100 kHz. That plateau sits exactly between 35 kHz and 138 kHz: the 11ac weight covers most of it, while the 11ax weight has already fallen to zero before the plateau ends.

For a plateau wider than $1/T$ there is a handy approximation, using

$$\int_0^\infty \text{sinc}^2(fT) df = \frac{1}{2T} \quad \Rightarrow \quad \sigma_{\text{CPE}}^2 \approx \frac{S_0}{2T}, \quad \text{removed fraction} \approx \frac{S_0}{2T\sigma_\phi^2} \propto \frac{1}{T}$$

With $S_0 = 7.8 \times 10^{-11}$ rad²/Hz and total phase power $\sigma_\phi^2 = 4.1 \times 10^{-5}$ rad² (10 kHz – 100 MHz, IPN -46.9 dBc):

	approx. $S_0/(2T)$	exact integral	CPE gain $-10 \cdot \log_{10}(1 - \text{fraction})$
11ac/n	29.6 %	27.8 %	1.41 dB
11ax/be	7.4 %	5.5 %	0.24 dB

Panel (3) is the cumulative exact integral: the green curve climbs to 27.8 %, the red one flattens around 35 kHz and stops at 5.5 %. This agrees with the time-domain model's direct readings of 1.55 / 0.36 dB (40 MHz, 32 frames); the difference is the integration band and realization spread.

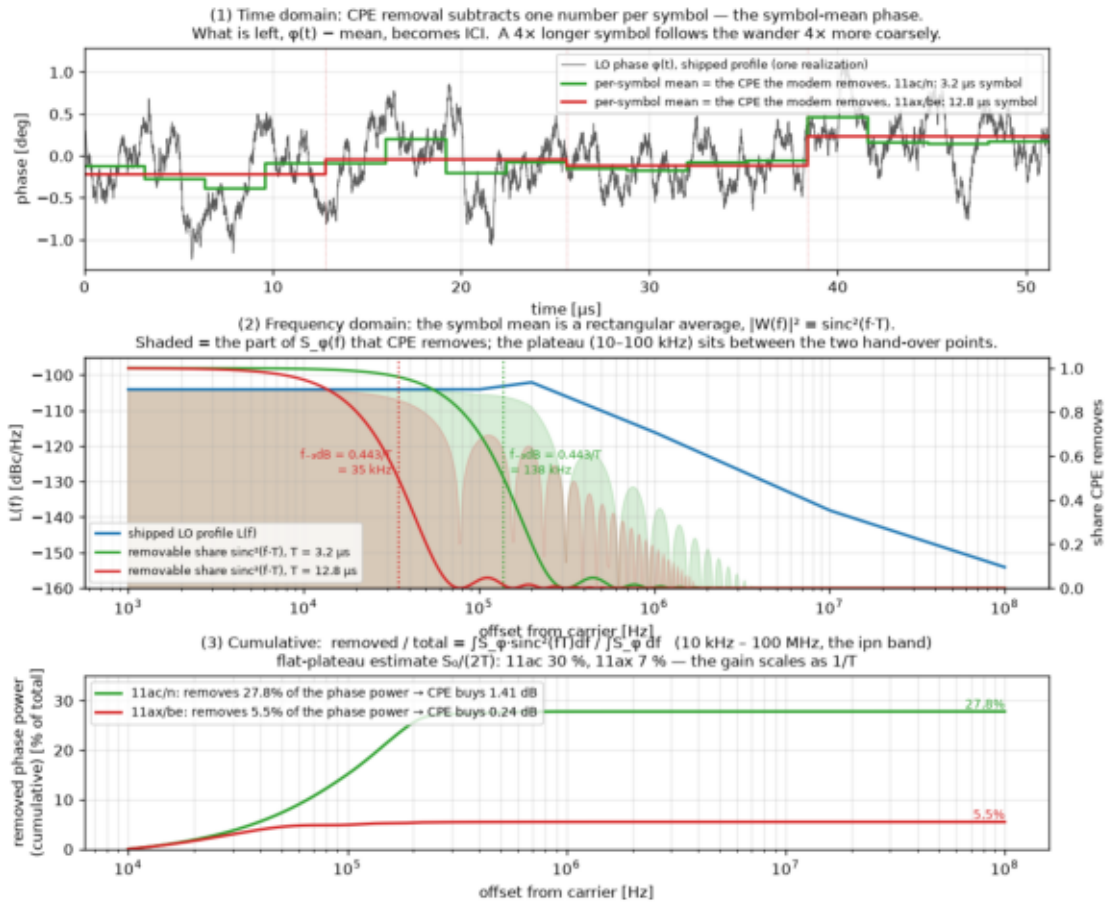
In one sentence

The CPE-removal gain is $\int S_\phi(f) \cdot \text{sinc}^2(fT) df$ and the weight's width scales as $1/T$. WiFi 7 stretches T by 4×, shrinking the removable band from 138 kHz to 35 kHz, and this class of PLL puts its noise plateau exactly between those two numbers — so 11ac takes out nearly a third of the phase power and 11ax less than a tenth. The extra loss is not a weaker modem algorithm; it is ICI, and it cannot be removed.

Where the numbers come from

Model: wifitrx pn_cpe_study (isolation method, phase noise the only impairment, true channel unity). Closed forms: cpe_partition() over the LO profile with the 12.8 μs / 3.2 μs FFT lengths, computed by this script. Time-domain readings: 40 MHz, 6-pilot CPE + LTF channel estimate, 32 frames; the standards gap in the modem form is 1.4 ± 0.2 dB (1σ over seeds), the ICI floor alone scatters 0.05 dB.

Figure: time domain, sinc² weight, cumulative removed share



(1) One LO phase realization with the per-symbol means for both symbol lengths. (2) $L(f)$ of the shipped profile with the removable share $\text{sinc}^2(fT)$ for $T = 3.2 \mu\text{s}$ and $12.8 \mu\text{s}$; shaded = $S_{\phi} \cdot \text{sinc}^2$. (3) Cumulative removed fraction over 10 kHz – 100 MHz: 27.8 % (11ac/n) vs 5.5 % (11ax/be).